Refresh dos ambientes OHI - Qualicorp

Os ambientes OHI da Qualicorp sofrem constantes refreshs.
Todos essas atualizações devem ser feitas via RMAN e não devem ser feitas via datapump, pois ocorrem diversos problemas.

Todos os bancos de dados estão em ambiente RAC de 2 nós.

Os servidores KAZUMBA/KABONG são destinados a UAT/DEV e contém os seguintes bancos de dados:

MACHML -> Ambiente MAC*
MACUAT -> Ambiente MAC*
OHICOD01
OHICOD02
OHICOD03
OHICOD04
OHICOD05
OHICOD06
OHIINT01
OHITRN01
OHITST01
OHIUAT01

*Os bancos de dados do MAC não sofrem atualizações como o OHI, caso essa demanda surja deve ser tratada pontualmente se será via RMAN ou DATAPUMP.

Para produção, temos os servidores QUIRINOPOLIS/FERNANDOPOLIS:

OHIPRD01 -> Produção
OHIMIG01 -> Banco de dados transitório, deve ser excluído em breve.

No MAC, os servidores de produção são AMPARO/LINDOIA:

MACPRD

Os passos descritos nesse documento, ilustram a atualização de um banco já existente e não a criação de um novo banco de dados.

As etapas necessárias são:
- Backup do banco de dados de Origem;
- Backup do banco de dados Destino (o que será atualizado);
- Transferência do backup (se necessário);
- Drop do banco de dados Destino;
- Acerto dos parâmetros no banco de dados Destino;
- Duplicate;
- Procedimento pós-restore (Documento da ACCENTURE)

Neste exemplo foi realizado o refresh do banco OHICOD02(Origem) para OHIPRD01(Destino).

Backup do banco de dados Origem

Neste caso, devemos realizar o backup do banco OHICOD02, os backups podem ser feitos com o banco de dados fechado (mount).

As áreas destinadas para esses backups são:
Nos servidores kazumba/kabong, utilizar o filesystem **/backup**
Nos servidores quirinopolis/fernandopolis, utilizar **/oracle/backup**

Execute os passos abaixo para a execução deste backup (pode ser executado em qualquer node do RAC):

- Crie um diretório para que o backup não seja misturado com os demais:

mkdir –p /backup/OHICOD02/BKP_20092013

- Setar o ORACLE_SID para a instance desejada:

Neste caso, estamos no servidor KAZUMBA, o SID do banco OHICOD02 será **OHICOD021:**
export ORACLE_SID=OHICOD021

- Baixar o banco de dados no cluster:

srvctl stop database -d OHICOD02

- Conectar no RMAN, montar o banco e rodar o backup:

Em seguida deve-se montar o banco, ele pode ser montado tanto pelo rman quanto pelo sqlplus.
Para conectar no rman digite:
rman target /

Monte o banco de dados:
startup mount

Execute o comando para executar o backup do banco, o parâmetro FORMAT informa onde será gerado o backup e qual será a nomenclatura, substitua a parte em vermelho para o diretório criado por você para o seu backup e a parte em verde pelo nome do seu banco. O backup de archives é opcional, deve ser executado somente nos bancos que estão em modo de archive:

Com archive:

```
RUN {
 ALLOCATE CHANNEL CH00 TYPE DISK;
 ALLOCATE CHANNEL CH01 TYPE DISK;
 ALLOCATE CHANNEL CH02 TYPE DISK;
 ALLOCATE CHANNEL CH03 TYPE DISK;
 BACKUP AS COMPRESSED BACKUPSET DATABASE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_db_%U.rman';
BACKUP AS COMPRESSED BACKUPSET ARCHIVELOG ALL FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_arch_%U.rman';
 BACKUP SPFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_spfile_%U.rman';
BACKUP CURRENT CONTROLFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_ctl_%U.rman';
}
```

Sem archive:

```
RUN {
 ALLOCATE CHANNEL CH00 TYPE DISK;
 ALLOCATE CHANNEL CH01 TYPE DISK;
 ALLOCATE CHANNEL CH02 TYPE DISK;
 ALLOCATE CHANNEL CH03 TYPE DISK;
 BACKUP AS COMPRESSED BACKUPSET DATABASE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_db_%U.rman';
BACKUP SPFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_spfile_%U.rman';
BACKUP CURRENT CONTROLFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_ctl_%U.rman';
}
```

A nomenclatura dos arquivos de backup é livre, porém da forma com que está sugerida fica fácil a identificação de cada tipo de arquivo.

- Subir o banco de dados no cluster:

Ao término com sucesso do backup, no prórpio prompt do RMAN baixe o banco de dados:
shutdown immediate

Sair do RMAN e subir o banco novamente pelo cluster:
srvctl start database -d OHICOD02

A execução desses comandos bem como suas saídas estão no APENDICE A no final deste documento.

Transferência dos arquivos para o servidor Destino

Este passo é opcional, visto que os bancos de origem e destino podem estar no mesmo servidor.
No nosso exemplo, o servidor destino é o OHIPRD01, portanto devem ser transferidos para quirinopolis/fernandopolis.

A transferência é feita via SCP, por se tratar de um cluster pode ser enviado para qualquer node.

Entrar no diretório onde os backups foram gerados:

```
[oradev@kazumba ~]$ cd /backup/OHICOD02/BKP_20092013/
[oradev@kazumba BKP_20092013]$ ls -l
total 7566556
-rw-r----- 1 oradev asmadmin   12419072 Sep 20 00:34 OHICOD02_ctl_08ok9hld_1_1.rman
-rw-r----- 1 oradev asmadmin 2087567360 Sep 20 00:34 OHICOD02_db_01ok9h9v_1_1.rman
-rw-r----- 1 oradev asmadmin 2114535424 Sep 20 00:34 OHICOD02_db_02ok9h9v_1_1.rman
-rw-r----- 1 oradev asmadmin 2142068736 Sep 20 00:34 OHICOD02_db_03ok9h9v_1_1.rman
-rw-r----- 1 oradev asmadmin 1382055936 Sep 20 00:31 OHICOD02_db_04ok9ha0_1_1.rman
-rw-r----- 1 oradev asmadmin    1146880 Sep 20 00:32 OHICOD02_db_05ok9hg3_1_1.rman
-rw-r----- 1 oradev asmadmin      98304 Sep 20 00:32 OHICOD02_db_06ok9hg5_1_1.rman
-rw-r----- 1 oradev asmadmin      98304 Sep 20 00:34 OHICOD02_spfile_07ok9hlc_1_1.rman
```

Transferir via scp os arquivos do backup, antes, criar um diretório para receber esses arquivos no servidor destino:

```
[oradev@kazumba BKP_20092013]$ scp * oraprd@quirinopolis:/oracle/backup/refresh_ohicod02
Warning: the RSA host key for 'quirinopolis' differs from the key for the IP address '172.22.35.73'
Offending key for IP in /home/oradev/.ssh/known_hosts:25
Matching host key in /home/oradev/.ssh/known_hosts:40
Are you sure you want to continue connecting (yes/no)? yes
oraprd@quirinopolis's password:
OHICOD02_ctl_08ok9hld_1_1.rman                100%   12MB  11.8MB/s   00:00
OHICOD02_db_01ok9h9v_1_1.rman                 100% 1991MB  76.6MB/s   00:26
OHICOD02_db_02ok9h9v_1_1.rman                 100% 2017MB  96.0MB/s   00:21
OHICOD02_db_03ok9h9v_1_1.rman                 100% 2043MB  92.9MB/s   00:22
OHICOD02_db_04ok9ha0_1_1.rman                 100% 1318MB  82.4MB/s   00:16
OHICOD02_db_05ok9hg3_1_1.rman                 100% 1120KB   1.1MB/s   00:00
OHICOD02_db_06ok9hg5_1_1.rman                 100%   96KB  96.0KB/s   00:00
OHICOD02_spfile_07ok9hlc_1_1.rman             100%   96KB  96.0KB/s   00:00
[oradev@kazumba BKP_20092013]$
```

Executar backup no banco Destino

Seguir os mesmos passos para realizar o backup do banco origem, adequando o diretório, ORACLE_SID e o nome do banco:

```
[oraprd@quirinopolis refresh_ohicod02]$ srvctl stop database -d ohiprd01
[oraprd@quirinopolis refresh_ohicod02]$ rman target /
RMAN> startup mount
RUN {
 ALLOCATE CHANNEL CH00 TYPE DISK;
 ALLOCATE CHANNEL CH01 TYPE DISK;
 ALLOCATE CHANNEL CH02 TYPE DISK;
 ALLOCATE CHANNEL CH03 TYPE DISK;
 BACKUP AS COMPRESSED BACKUPSET DATABASE FORMAT
'/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_%U.rman';
 BACKUP SPFILE FORMAT '/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_spfile_%U.rman';
 BACKUP CURRENT CONTROLFILE FORMAT '/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_ctl_%U.rman';
}
```

O banco de dados pode continuar em mount, não é necessário subi-lo novamente.

A saída completa desses comandos estão no APENDICE B.

Gerar script para dropar o banco

No banco de dados DESTINO, executar o select abaixo e salvar a saída, ele irá gerar os comandos para dropar todos os arquivos do banco de dados:

```
set lines 900 trims on pages 300
 select 'alter diskgroup ' ||substr(name,2,instr(name,'/')-2) || ' drop file ''' || name || ''';' from v$datafile
union
 select 'alter diskgroup ' ||substr(name,2,instr(name,'/')-2) || ' drop file ''' || name || ''';' from v$tempfile
 union
 select 'alter diskgroup ' ||substr(name,2,instr(name,'/')-2) || ' drop file ''' || name || ''';' from v$controlfile
 union
 select 'alter diskgroup ' ||substr(member,2,instr(member,'/')-2) || ' drop file ''' || member || ''';' from v$logfile;
```

Ajuste do banco para o duplicate (banco destino)

Os parâmetros abaixo são setados para garantir que os caminhos dos datafiles sejam alterados de acordo.
Não precisa ser passado os paths completo dos datafiles nem dos redologs, basta o começo dos caminhos para a conversão.
Os Diskgroups tem o mesmo nome nos ambientes, adéqüe o segundo diretório de acordo com a origem destino.

- Abaixo para a conversão dos datafiles;

alter system set db_file_name_convert='+OHIDATA/BD_ORIGEM','+OHIDATA/BD_DESTINO' scope=spfile sid='*';

Exemplo:
alter system set db_file_name_convert='+OHIDATA/ohicod02','+OHIDATA/ohiprd01' scope=spfile sid='*';

Para alterar os redologs, execute abaixo, como há dois destinos para redologs, os dois devem ser passados:

alter system set log_file_name_convert='+OHIREDO/BD_ORIGEM','+OHIREDO/BD_DESTINO','+OHIRECO/BD_ORIGEM','+OHIRECO/BD_DESTINO' scope=spfile sid='*';

Exemplo:
alter system set log_file_name_convert='+OHIREDO/ohicod02','+OHIREDO/ohiprd01','+OHIRECO/ohicod02','+OHIRECO/ohiprd01' scope=spfile sid='*';

- Desabilitar o cluster do banco, toda e qualquer atividade de restore requer que esse parametro esteja como FALSE, do contrário dará falha:

alter system set cluster_database=false sid='*' scope=spfile;

- Verificar se o banco de dados destino está em modo archive, pois ao término da atividade deverá ser habilitado/desabilitado para que fique de acordo, no sqlplus:

Archive log list

- Baixar o banco de dados, subir em nomount:

shutdown immediate
startup nomount

- Dropar o banco de dados:

Executar, via sqlplus, os comandos para dropar os arquivos salvos anteriormente.

Executar o duplicate

No banco destino, conectar no RMAN como AUXILIARY:
rman auxiliary /

Executar o comando duplicate, o caminho em vermelho deve ser o local para onde foram trasnferidos os arquivos de backup do banco origem e, o nome do banco em verde deve ser o banco destino:

duplicate database to OHIPRD01 backup location '/oracle/backup/refresh_ohicod02/' nofilenamecheck;

A saída completa do comando está no APENDICE C.

Ao término do duplicate, o banco estará OPEN.

- Ajustar o archivelog:

No nosso caso, o banco de dados OHIPRD01 estava em modo de archive e o banco OHICOD02 não. Portanto, ao término do duplicate o banco OHIPRD01 não está em modo archive, isso deve ser corrigido passando ele para archive. Neste momento o banco já pode voltar para o cluster:

SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup mount
ORACLE instance started.

Total System Global Area 3.1868E+10 bytes
Fixed Size 2244072 bytes
Variable Size 4294967832 bytes
Database Buffers 2.7515E+10 bytes
Redo Buffers 56164352 bytes
Database mounted.
SQL> alter database archivelog;

Database altered.

SQL> show parameter cluster

NAME TYPE VALUE
------------------------------------ ----------- ------------------------------
cluster_database boolean FALSE
cluster_database_instances integer 1
cluster_interconnects string
SQL> alter system set cluster_database=true sid='*' scope=spfile;

System altered.

SQL> shutdown immediate
ORA-01109: database not open

Database dismounted.
ORACLE instance shut down.
SQL> exit
Disconnected from Oracle Database 11g Enterprise Edition Release 11.2.0.3.0 - 64bit Production
With the Partitioning, Real Application Clusters, Automatic Storage Management, OLAP,
Data Mining and Real Application Testing options

[oraprd@quirinopolis refresh_ohicod02]$ srvctl start database -d ohiprd01
[oraprd@quirinopolis refresh_ohicod02]$ srvctl status database -d ohiprd01
Instance OHIPRD01B is running on node fernandopolis
Instance OHIPRD01A is running on node quirinopolis

Procedimento pós-restore

Esses procedimentos estão descritos no document
"Projeto_NMA_Procedimento_pos_backup_v2.docx" enviado pela ACCENTURE.
Ele envolve passos necessários para ajustar o banco de dados para a aplicação funcionar corretamente.

Os passos abaixo são apenas um copy/paste do documento, comentários meus estarão em vermelho.
Os valores devem ser ajustados de acordo com o banco destino.

Ajustes na base de dados

Para a aplicação OHI conseguir logar na base de dados, é necessário atualizar dentro da base o IP do servidor de aplicação. Como a base do exemplo acima é uma copia do ambiente OHICOD02, dentro desta base está gravado o IP do servidor de aplicação do ambiente OHICOD02 (servidor soacha). Assim, neste exemplo é necessário remover estes dois endereços IPs e incluir os dois endereços IPs dos servidores do ambiente OHIPRD01 (Macas e Mantas). Para isso seguir os passos abaixo:

1. Logar na base de dados
    a. sqlplus ozg_owner/ozg_owner@$ORACLE_SID
    b. Se falhar, sqlplus ozg_owner/ozg_owner9@$ORACLE_SID
        i. $ORACLE_SID = nome do ambiente (Ex:. OHIINT01, OHIUAT01, etc)

2. Remover o IP dos servidores de aplicação antigos (OHICOD02)
    a. exec alg_ias_pck.del('172.22.33.31');
        i. IP SOACHA: 172.22.33.31

Para saber os valores a serem deletados, execute a query:

select * from alg#ip_adressen;

3. Incluir o IP dos servidores de aplicação novos (OHIPRD01)
    a. exec alg_ias_pck.ins('172.22.38.131');
    b. exec alg_ias_pck.ins('172.22.38.132');

4. Comitar mudanças
    a. commit;

5. Sair do banco de dados
    a. Exit

Por fim, é necessário atualizar o valor da variável OZG_TMP que está inserida dentro do banco de dados. A relação completa dos valores desta variável por ambiente pode ser encontrado no **Anexo A.** Seguem instruções para atualizar a variável:

1. Logar no banco com o usuário SYS
    a. sqlplus sys/Oracle123@$ORACLE_SID as sysdba
2. Atualizar o valor da variável OZG_TMP, utilizando o valor indicado na tabela do **Anexo A.**
    a. create or replace directory ozg_tmp as '/u01/oracle/ohidata_tst/OHIINT01/tmp';
3. Executar como SYS, o script anexo para atualização das demais variáveis de ambiente. Utilizar na chamada do script o

parâmetro: [nome do servidor master de aplicação]. Por exemplo, se o refresh da base está sendo feito do ambiente UAT01 para o ambiente INT01, o nome do servidor de aplicação destino é o **tulia** (caixa baixa). Assim a chamada seria feita da forma:

a. @alter_system_parameter.sql **tulia**
    i. Para a lista completa de nome de servidores masters, consultar o **Anexo A.**

Anexo A

Tabela com o nome, endereço dos servidores de aplicação OHI

Módulo	Environment	Server Master	Descrição	IP	Usuário APP	OZG_TMP
OHI	DEV	ibague	COD01	172.22.33.30	oradev	/u01/oracle/ohidata_dev/OHICOD01/tmp
		soacha	COD02	172.22.33.31		/u01/oracle/ohidata_dev/OHICOD02/tmp
		pasto	COD03	172.22.33.32		/u01/oracle/ohidata_dev/OHICOD03/tmp
		monteria	COD04	172.22.33.33		/u01/oracle/ohidata_dev/OHICOD04/tmp
		malambo	COD05	172.22.33.38		/u01/oracle/ohidata_dev/OHICOD05/tmp
		zipaguira	COD06	172.22.33.39		/u01/oracle/ohidata_dev/OHICOD06/tmp
	TST	sincelejo	TST01	172.22.33.34		/u01/oracle/ohidata_dev/OHITST01/tmp
	TRN	quibdo	TRN01	172.22.33.35		/u01/oracle/ohidata_dev/OHITRN01/tmp
	INT	tulia	INT01	172.22.33.36		/u01/oracle/ohidata_tst/OHIINT01/tmp
				172.22.33.37		
	UAT	girardot	UAT01	172.22.34.30	orauat	/u01/oracle/ohidata_uat/OHIUAT01/tmp
				172.22.34.31		
	PRD	macas	PRD01	172.22.38.131	oraprd	/u01/oracle/ohidata/OHIPRD01/tmp
				172.22.38.132		

APENDICE A

[oradev@kazumba ~]$ export ORACLE_SID=OHICOD021
[oradev@kazumba ~]$ srvctl stop database -d OHICOD02
[oradev@kazumba ~]$ rman target /

Recovery Manager: Release 11.2.0.3.0 - Production on Fri Sep 20 00:27:32 2013

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connected to target database (not started)

RMAN> startup mount

Oracle instance started
database mounted

Total System Global Area 4075356160 bytes

Fixed Size 2234720 bytes
Variable Size 1509951136 bytes
Database Buffers 2550136832 bytes
Redo Buffers 13033472 bytes
RMAN> **RUN {**
2> **ALLOCATE CHANNEL CH00 TYPE DISK;**
3> **ALLOCATE CHANNEL CH01 TYPE DISK;**
4> **ALLOCATE CHANNEL CH02 TYPE DISK;**
5> **ALLOCATE CHANNEL CH03 TYPE DISK;**
6> **BACKUP AS COMPRESSED BACKUPSET DATABASE FORMAT**
'/backup/OHICOD02/BKP_20092013/OHICOD02_db_%U.rman';
7> **BACKUP SPFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_spfile_%U.rman';**
BACKUP CURRENT CONTROLFILE FORMAT '/backup/OHICOD02/BKP_20092013/OHICOD02_ctl_%U.rman';
8> 9> **}**

using target database control file instead of recovery catalog
allocated channel: CH00
channel CH00: SID=3 instance=OHICOD021 device type=DISK

allocated channel: CH01
channel CH01: SID=98 instance=OHICOD021 device type=DISK

allocated channel: CH02
channel CH02: SID=142 instance=OHICOD021 device type=DISK

allocated channel: CH03
channel CH03: SID=187 instance=OHICOD021 device type=DISK

Starting backup at 20-SEP-13
channel CH00: starting compressed full datafile backup set
channel CH00: specifying datafile(s) in backup set
input datafile file number=00025 name=+OHIDATA/ohicod02/datafile/tbs_control_01.dbf
input datafile file number=00020 name=+OHIDATA/ohicod02/datafile/ozg_fact_zrg_tab.341.825854355
input datafile file number=00010 name=+OHIDATA/ohicod02/datafile/ozg_dim_sys_tab.600.825854351
input datafile file number=00007 name=+OHIDATA/ohicod02/datafile/ozg_dim_rel_ind.348.825854353

input datafile file number=00012 name=+OHIDATA/ohicod02/datafile/ozg_dim_zrg_tab.551.825854353
input datafile file number=00022 name=+OHIDATA/ohicod02/datafile/ozg_log_tab.345.825854355
channel CH00: starting piece 1 at 20-SEP-13
channel CH01: starting compressed full datafile backup set
channel CH01: specifying datafile(s) in backup set
input datafile file number=00026 name=+OHIDATA/ohicod02/datafile/tbs_control_02.dbf
input datafile file number=00017 name=+OHIDATA/ohicod02/datafile/ozg_fact_sys_ind.603.825854351
input datafile file number=00009 name=+OHIDATA/ohicod02/datafile/ozg_dim_sys_ind.608.825854351
input datafile file number=00015 name=+OHIDATA/ohicod02/datafile/ozg_fact_rel_ind.339.825854353
input datafile file number=00006 name=+OHIDATA/ohicod02/datafile/ozg_dim_fin_tab.377.825854353
input datafile file number=00011 name=+OHIDATA/ohicod02/datafile/ozg_dim_zrg_ind.594.825854353
input datafile file number=00021 name=+OHIDATA/ohicod02/datafile/ozg_log_ind.331.825854355
channel CH01: starting piece 1 at 20-SEP-13
channel CH02: starting compressed full datafile backup set
channel CH02: specifying datafile(s) in backup set
input datafile file number=00024 name=+OHIDATA/ohicod02/datafile/tbs_control.dbf
input datafile file number=00019 name=+OHIDATA/ohicod02/datafile/ozg_fact_zrg_ind.344.825854355
input datafile file number=00004 name=+OHIDATA/ohicod02/datafile/users.343.825854353
input datafile file number=00005 name=+OHIDATA/ohicod02/datafile/ozg_dim_fin_ind.607.825854353
input datafile file number=00008 name=+OHIDATA/ohicod02/datafile/ozg_dim_rel_tab.332.825854353
input datafile file number=00016 name=+OHIDATA/ohicod02/datafile/ozg_fact_rel_tab.340.825854353
channel CH02: starting piece 1 at 20-SEP-13
channel CH03: starting compressed full datafile backup set
channel CH03: specifying datafile(s) in backup set
input datafile file number=00003 name=+OHIDATA/ohicod02/datafile/undotbs1.597.825854347
input datafile file number=00001 name=+OHIDATA/ohicod02/datafile/system.610.825854347
input datafile file number=00018 name=+OHIDATA/ohicod02/datafile/ozg_fact_sys_tab.591.825854347
input datafile file number=00002 name=+OHIDATA/ohicod02/datafile/sysaux.599.825854347
input datafile file number=00014 name=+OHIDATA/ohicod02/datafile/ozg_fact_fin_tab.602.825854353
input datafile file number=00023 name=+OHIDATA/ohicod02/datafile/undotbs2.385.825854351
input datafile file number=00013 name=+OHIDATA/ohicod02/datafile/ozg_fact_fin_ind.595.825854353
channel CH03: starting piece 1 at 20-SEP-13
channel CH03: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_04ok9ha0_1_1.rman tag=TAG20130920T002846 comment=NONE
channel CH03: backup set complete, elapsed time: 00:03:15
channel CH03: starting compressed full datafile backup set
channel CH03: specifying datafile(s) in backup set
including current control file in backup set
channel CH03: starting piece 1 at 20-SEP-13
channel CH03: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_05ok9hg3_1_1.rman tag=TAG20130920T002846 comment=NONE
channel CH03: backup set complete, elapsed time: 00:00:01
channel CH03: starting compressed full datafile backup set
channel CH03: specifying datafile(s) in backup set
including current SPFILE in backup set
channel CH03: starting piece 1 at 20-SEP-13
channel CH03: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_06ok9hg5_1_1.rman tag=TAG20130920T002846 comment=NONE
channel CH03: backup set complete, elapsed time: 00:00:01
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_01ok9h9v_1_1.rman tag=TAG20130920T002846 comment=NONE

channel CH00: backup set complete, elapsed time: 00:05:53
channel CH02: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_03ok9h9v_1_1.rman tag=TAG20130920T002846 comment=NONE
channel CH02: backup set complete, elapsed time: 00:05:52
channel CH01: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_db_02ok9h9v_1_1.rman tag=TAG20130920T002846 comment=NONE
channel CH01: backup set complete, elapsed time: 00:06:03
Finished backup at 20-SEP-13

Starting backup at 20-SEP-13
channel CH00: starting full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current SPFILE in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_spfile_07ok9hlc_1_1.rman tag=TAG20130920T003452 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
Finished backup at 20-SEP-13

Starting backup at 20-SEP-13
channel CH00: starting full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current control file in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/backup/OHICOD02/BKP_20092013/OHICOD02_ctl_08ok9hld_1_1.rman tag=TAG20130920T003453 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
Finished backup at 20-SEP-13
released channel: CH00
released channel: CH01
released channel: CH02
released channel: CH03

RMAN> shutdown immediate

database dismounted
Oracle instance shut down

RMAN> exit

Recovery Manager complete.

[oradev@kazumba ~]$ srvctl start database -d OHICOD02
[oradev@kazumba ~]$ srvctl status database -d OHICOD02
Instance OHICOD022 is running on node kabong
Instance OHICOD021 is running on node kazumba
[oradev@kazumba ~]$ sqlplus

SQL*Plus: Release 11.2.0.3.0 Production on Fri Sep 20 00:46:45 2013

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```
Enter user-name: / as sysdba

Connected to:
Oracle Database 11g Enterprise Edition Release 11.2.0.3.0 - 64bit Production
With the Partitioning, Real Application Clusters, Automatic Storage Management, Data Mining
and Real Application Testing options

SQL> select host_name, instance_name, status from gv$instance;

HOST_NAME
----------------------------------------------------------------
INSTANCE_NAME    STATUS
---------------- ------------
kazumba
OHICOD021        OPEN

kabong
OHICOD022        OPEN


SQL> exit
Disconnected from Oracle Database 11g Enterprise Edition Release 11.2.0.3.0 - 64bit Production
With the Partitioning, Real Application Clusters, Automatic Storage Management, Data Mining
and Real Application Testing options
```

APENDICE B

[oraprd@quirinopolis refresh_ohicod02]$ srvctl stop database -d ohiprd01
[oraprd@quirinopolis refresh_ohicod02]$ rman target /

Recovery Manager: Release 11.2.0.3.0 - Production on Fri Sep 20 00:53:08 2013

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connected to target database (not started)

RMAN> startup mount

Oracle instance started
database mounted

Total System Global Area 31868010496 bytes

Fixed Size 2244072 bytes
Variable Size 4294967832 bytes
Database Buffers 27514634240 bytes
Redo Buffers 56164352 bytes

RMAN> **RUN {**
2> **ALLOCATE CHANNEL CH00 TYPE DISK;**
3> **ALLOCATE CHANNEL CH01 TYPE DISK;**
4> **ALLOCATE CHANNEL CH02 TYPE DISK;**
5> **ALLOCATE CHANNEL CH03 TYPE DISK;**
6> **BACKUP AS COMPRESSED BACKUPSET DATABASE FORMAT**
'/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_%U.rman';
7> **BACKUP SPFILE FORMAT '/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_spfile_%U.rman';**
8> **BACKUP CURRENT CONTROLFILE FORMAT**
'/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_ctl_%U.rman';
}9>

using target database control file instead of recovery catalog
allocated channel: CH00
channel CH00: SID=3 instance=OHIPRD01A device type=DISK

allocated channel: CH01
channel CH01: SID=98 instance=OHIPRD01A device type=DISK

allocated channel: CH02
channel CH02: SID=142 instance=OHIPRD01A device type=DISK

allocated channel: CH03
channel CH03: SID=187 instance=OHIPRD01A device type=DISK

Starting backup at 20-SEP-13
channel CH00: starting compressed full datafile backup set
channel CH00: specifying datafile(s) in backup set
input datafile file number=00003 name=+OHIDATA/ohiprd01/datafile/undotbs1.279.826371335
input datafile file number=00019 name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_ind.281.826371337
input datafile file number=00020 name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_tab.266.826371337

input datafile file number=00021 name=+OHIDATA/ohiprd01/datafile/ozg_log_ind.262.826371337
input datafile file number=00022 name=+OHIDATA/ohiprd01/datafile/ozg_log_tab.276.826371337
channel CH00: starting piece 1 at 20-SEP-13
channel CH01: starting compressed full datafile backup set
channel CH01: specifying datafile(s) in backup set
input datafile file number=00001 name=+OHIDATA/ohiprd01/datafile/system.274.826371335
input datafile file number=00013 name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_ind.261.826371337
input datafile file number=00014 name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_tab.260.826371337
input datafile file number=00015 name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_ind.286.826371337
input datafile file number=00016 name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_tab.265.826371337
input datafile file number=00004 name=+OHIDATA/ohiprd01/datafile/users.259.826371337
channel CH01: starting piece 1 at 20-SEP-13
channel CH02: starting compressed full datafile backup set
channel CH02: specifying datafile(s) in backup set
input datafile file number=00002 name=+OHIDATA/ohiprd01/datafile/sysaux.285.826371335
input datafile file number=00023 name=+OHIDATA/ohiprd01/datafile/undotbs2.277.826371335
input datafile file number=00007 name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_ind.273.826371337
input datafile file number=00008 name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_tab.275.826371337
input datafile file number=00011 name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_ind.267.826371337
input datafile file number=00012 name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_tab.268.826371337
channel CH02: starting piece 1 at 20-SEP-13
channel CH03: starting compressed full datafile backup set
channel CH03: specifying datafile(s) in backup set
input datafile file number=00018 name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_tab.283.826371335
input datafile file number=00017 name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_ind.264.826371335
input datafile file number=00009 name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_ind.278.826371337
input datafile file number=00010 name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_tab.284.826371337
input datafile file number=00005 name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_ind.271.826371337
input datafile file number=00006 name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_tab.257.826371337
channel CH03: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1cok9ioh_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:03
channel CH00: starting compressed full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current control file in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1gok9iok_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
channel CH00: starting compressed full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current SPFILE in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1hok9ion_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
channel CH02: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1eok9ioh_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH02: backup set complete, elapsed time: 00:01:00
channel CH03: finished piece 1 at 20-SEP-13

```
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1fok9ioi_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH03: backup set complete, elapsed time: 00:01:49
channel CH01: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_db_1dok9ioh_1_1.rman
tag=TAG20130920T005336 comment=NONE
channel CH01: backup set complete, elapsed time: 00:03:11
Finished backup at 20-SEP-13

Starting backup at 20-SEP-13
channel CH00: starting full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current SPFILE in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_spfile_1iok9iuh_1_1.rman
tag=TAG20130920T005649 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
Finished backup at 20-SEP-13

Starting backup at 20-SEP-13
channel CH00: starting full datafile backup set
channel CH00: specifying datafile(s) in backup set
including current control file in backup set
channel CH00: starting piece 1 at 20-SEP-13
channel CH00: finished piece 1 at 20-SEP-13
piece handle=/oracle/backup/OHIPRD01/BKPFULL/OHIPRD01_cold_ctl_1jok9iui_1_1.rman
tag=TAG20130920T005650 comment=NONE
channel CH00: backup set complete, elapsed time: 00:00:01
Finished backup at 20-SEP-13
released channel: CH00
released channel: CH01
released channel: CH02
released channel: CH03

RMAN>



RMAN> exit


Recovery Manager complete.
```

APENDICE C

[oraprd@quirinopolis refresh_ohicod02]$ rman auxiliary /

Recovery Manager: Release 11.2.0.3.0 - Production on Fri Sep 20 01:01:19 2013

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connected to auxiliary database: OHIPRD01 (not mounted)

RMAN> duplicate database to OHIPRD01 backup location '/oracle/backup/refresh_ohicod02/' nofilenamecheck;

Starting Duplicate Db at 20-SEP-13

contents of Memory Script:
{
  sql clone "alter system set control_files =
 ''+OHIDATA/ohiprd01/controlfile/current.270.826592505'',
''+OHIRECO/ohiprd01/controlfile/current.306.826592505'' comment=
 ''Set by RMAN'' scope=spfile";
  sql clone "alter system set db_name =
 ''OHICOD02'' comment=
 ''Modified by RMAN duplicate'' scope=spfile";
  sql clone "alter system set db_unique_name =
 ''OHIPRD01'' comment=
 ''Modified by RMAN duplicate'' scope=spfile";
  shutdown clone immediate;
  startup clone force nomount
  restore clone primary controlfile from '/oracle/backup/refresh_ohicod02/OHICOD02_ctl_08ok9hld_1_1.rman';
  alter clone database mount;
}
executing Memory Script

sql statement: alter system set control_files = ''+OHIDATA/ohiprd01/controlfile/current.270.826592505'',
''+OHIRECO/ohiprd01/controlfile/current.306.826592505'' comment= ''Set by RMAN'' scope=spfile

sql statement: alter system set db_name = ''OHICOD02'' comment= ''Modified by RMAN duplicate'' scope=spfile

sql statement: alter system set db_unique_name = ''OHIPRD01'' comment= ''Modified by RMAN duplicate''
scope=spfile

Oracle instance shut down

Oracle instance started

Total System Global Area   31868010496 bytes

Fixed Size   2244072 bytes
Variable Size   4294967832 bytes
Database Buffers   27514634240 bytes
Redo Buffers   56164352 bytes

Starting restore at 20-SEP-13
allocated channel: ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: SID=371 device type=DISK

channel ORA_AUX_DISK_1: restoring control file
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:05
output file name=+OHIDATA/ohiprd01/controlfile/current.270.826592505
output file name=+OHIRECO/ohiprd01/controlfile/current.306.826592505
Finished restore at 20-SEP-13

database mounted
released channel: ORA_AUX_DISK_1
allocated channel: ORA_AUX_DISK_1
channel ORA_AUX_DISK_1: SID=371 device type=DISK
RMAN-05529: WARNING: DB_FILE_NAME_CONVERT resulted in invalid ASM names; names changed to disk group only.

contents of Memory Script:
```
{
  set newname for datafile  1 to
 "+ohidata";
  set newname for datafile  2 to
 "+ohidata";
  set newname for datafile  3 to
 "+ohidata";
  set newname for datafile  4 to
 "+ohidata";
  set newname for datafile  5 to
 "+ohidata";
  set newname for datafile  6 to
 "+ohidata";
  set newname for datafile  7 to
 "+ohidata";
  set newname for datafile  8 to
 "+ohidata";
  set newname for datafile  9 to
 "+ohidata";
  set newname for datafile  10 to
 "+ohidata";
  set newname for datafile  11 to
 "+ohidata";
  set newname for datafile  12 to
 "+ohidata";
  set newname for datafile  13 to
 "+ohidata";
  set newname for datafile  14 to
 "+ohidata";
  set newname for datafile  15 to
 "+ohidata";
  set newname for datafile  16 to
 "+ohidata";
  set newname for datafile  17 to
 "+ohidata";
  set newname for datafile  18 to
 "+ohidata";
  set newname for datafile  19 to
 "+ohidata";
```

```
  set newname for datafile  20 to
 "+ohidata";
  set newname for datafile  21 to
 "+ohidata";
  set newname for datafile  22 to
 "+ohidata";
  set newname for datafile  23 to
 "+ohidata";
  set newname for datafile  24 to
 "+OHIDATA/ohiprd01/datafile/tbs_control.dbf";
  set newname for datafile  25 to
 "+OHIDATA/ohiprd01/datafile/tbs_control_01.dbf";
  set newname for datafile  26 to
 "+OHIDATA/ohiprd01/datafile/tbs_control_02.dbf";
  restore
  clone database
  ;
}
executing Memory Script

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

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executing command: SET NEWNAME

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executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

Starting restore at 20-SEP-13
using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00001 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00002 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00003 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00013 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00014 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00018 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00023 to +ohidata
channel ORA_AUX_DISK_1: reading from backup piece
/oracle/backup/refresh_ohicod02/OHICOD02_db_04ok9ha0_1_1.rman

channel ORA_AUX_DISK_1: piece handle=/oracle/backup/refresh_ohicod02/OHICOD02_db_04ok9ha0_1_1.rman
tag=TAG20130920T002846
channel ORA_AUX_DISK_1: restored backup piece 1
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:06:05
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00004 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00005 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00008 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00016 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00019 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00024 to +OHIDATA/ohiprd01/datafile/tbs_control.dbf
channel ORA_AUX_DISK_1: reading from backup piece
/oracle/backup/refresh_ohicod02/OHICOD02_db_03ok9h9v_1_1.rman

channel ORA_AUX_DISK_1: piece handle=/oracle/backup/refresh_ohicod02/OHICOD02_db_03ok9h9v_1_1.rman
tag=TAG20130920T002846
channel ORA_AUX_DISK_1: restored backup piece 1
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:08:25
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00006 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00009 to +ohidata

channel ORA_AUX_DISK_1: restoring datafile 00011 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00015 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00017 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00021 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00026 to +OHIDATA/ohiprd01/datafile/tbs_control_02.dbf
channel ORA_AUX_DISK_1: reading from backup piece
/oracle/backup/refresh_ohicod02/OHICOD02_db_02ok9h9v_1_1.rman

channel ORA_AUX_DISK_1: piece handle=/oracle/backup/refresh_ohicod02/OHICOD02_db_02ok9h9v_1_1.rman
tag=TAG20130920T002846
channel ORA_AUX_DISK_1: restored backup piece 1
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:08:36
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00007 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00010 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00012 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00020 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00022 to +ohidata
channel ORA_AUX_DISK_1: restoring datafile 00025 to +OHIDATA/ohiprd01/datafile/tbs_control_01.dbf
channel ORA_AUX_DISK_1: reading from backup piece
/oracle/backup/refresh_ohicod02/OHICOD02_db_01ok9h9v_1_1.rman

channel ORA_AUX_DISK_1: piece handle=/oracle/backup/refresh_ohicod02/OHICOD02_db_01ok9h9v_1_1.rman
tag=TAG20130920T002846
channel ORA_AUX_DISK_1: restored backup piece 1
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:08:45
Finished restore at 20-SEP-13

contents of Memory Script:
{
  switch clone datafile all;
}
executing Memory Script

datafile 1 switched to datafile copy
input datafile copy RECID=27 STAMP=826594446 file name=+OHIDATA/ohiprd01/datafile/system.277.826592537
datafile 2 switched to datafile copy
input datafile copy RECID=28 STAMP=826594446 file name=+OHIDATA/ohiprd01/datafile/sysaux.274.826592537
datafile 3 switched to datafile copy
input datafile copy RECID=29 STAMP=826594446 file name=+OHIDATA/ohiprd01/datafile/undotbs1.259.826592535
datafile 4 switched to datafile copy
input datafile copy RECID=30 STAMP=826594447 file name=+OHIDATA/ohiprd01/datafile/users.286.826592905
datafile 5 switched to datafile copy
input datafile copy RECID=31 STAMP=826594447 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_ind.283.826592905
datafile 6 switched to datafile copy
input datafile copy RECID=32 STAMP=826594447 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_tab.284.826593409
datafile 7 switched to datafile copy
input datafile copy RECID=33 STAMP=826594447 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_ind.280.826593925
datafile 8 switched to datafile copy

input datafile copy RECID=34 STAMP=826594447 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_tab.264.826592905
datafile 9 switched to datafile copy
input datafile copy RECID=35 STAMP=826594448 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_ind.268.826593409
datafile 10 switched to datafile copy
input datafile copy RECID=36 STAMP=826594448 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_tab.271.826593925
datafile 11 switched to datafile copy
input datafile copy RECID=37 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_ind.278.826593411
datafile 12 switched to datafile copy
input datafile copy RECID=38 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_tab.272.826593925
datafile 13 switched to datafile copy
input datafile copy RECID=39 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_ind.262.826592539
datafile 14 switched to datafile copy
input datafile copy RECID=40 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_tab.285.826592539
datafile 15 switched to datafile copy
input datafile copy RECID=41 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_ind.267.826593409
datafile 16 switched to datafile copy
input datafile copy RECID=42 STAMP=826594449 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_tab.265.826592905
datafile 17 switched to datafile copy
input datafile copy RECID=43 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_ind.261.826593409
datafile 18 switched to datafile copy
input datafile copy RECID=44 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_tab.279.826592537
datafile 19 switched to datafile copy
input datafile copy RECID=45 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_ind.281.826592905
datafile 20 switched to datafile copy
input datafile copy RECID=46 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_tab.257.826593925
datafile 21 switched to datafile copy
input datafile copy RECID=47 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_log_ind.275.826593411
datafile 22 switched to datafile copy
input datafile copy RECID=48 STAMP=826594450 file
name=+OHIDATA/ohiprd01/datafile/ozg_log_tab.282.826593925
datafile 23 switched to datafile copy
input datafile copy RECID=49 STAMP=826594450 file name=+OHIDATA/ohiprd01/datafile/undotbs2.276.826592539
datafile 24 switched to datafile copy
input datafile copy RECID=50 STAMP=826594451 file name=+OHIDATA/ohiprd01/datafile/tbs_control.dbf
datafile 25 switched to datafile copy
input datafile copy RECID=51 STAMP=826594451 file name=+OHIDATA/ohiprd01/datafile/tbs_control_01.dbf
datafile 26 switched to datafile copy
input datafile copy RECID=52 STAMP=826594451 file name=+OHIDATA/ohiprd01/datafile/tbs_control_02.dbf

contents of Memory Script:
{

```
  recover
  clone database
  noredo
   delete archivelog
  ;
}
executing Memory Script

Starting recover at 20-SEP-13
using channel ORA_AUX_DISK_1

Finished recover at 20-SEP-13
Oracle instance started

Total System Global Area   31868010496 bytes

Fixed Size                  2244072 bytes
Variable Size            4294967832 bytes
Database Buffers         27514634240 bytes
Redo Buffers                56164352 bytes

contents of Memory Script:
{
  sql clone "alter system set  db_name =
 ''OHIPRD01'' comment=
 ''Reset to original value by RMAN'' scope=spfile";
  sql clone "alter system reset  db_unique_name scope=spfile";
  shutdown clone immediate;
  startup clone nomount;
}
executing Memory Script

sql statement: alter system set  db_name =  ''OHIPRD01'' comment= ''Reset to original value by RMAN'' scope=spfile

sql statement: alter system reset  db_unique_name scope=spfile

Oracle instance shut down

connected to auxiliary database (not started)
Oracle instance started

Total System Global Area   31868010496 bytes

Fixed Size                  2244072 bytes
Variable Size            4294967832 bytes
Database Buffers         27514634240 bytes
Redo Buffers                56164352 bytes
sql statement: CREATE CONTROLFILE REUSE SET DATABASE "OHIPRD01" RESETLOGS NOARCHIVELOG
  MAXLOGFILES     16
  MAXLOGMEMBERS      3
  MAXDATAFILES      100
  MAXINSTANCES     8
  MAXLOGHISTORY     2045
 LOGFILE
  GROUP   1 ( '+ohireco', '+ohiredo' ) SIZE 50 M  REUSE,
```

```
  GROUP   2 ( '+ohireco', '+ohiredo' ) SIZE 50 M  REUSE
 DATAFILE
  '+OHIDATA/ohiprd01/datafile/system.277.826592537'
 CHARACTER SET AL32UTF8

sql statement: ALTER DATABASE ADD LOGFILE

  INSTANCE 'i2'
  GROUP   3 ( '+ohiredo', '+ohireco' ) SIZE 50 M  REUSE,
  GROUP   4 ( '+ohiredo', '+ohireco' ) SIZE 50 M  REUSE

contents of Memory Script:
{
   set newname for tempfile  1 to
 "+ohidata";
   set newname for tempfile  2 to
 "+OHIDATA/ohiprd01/tempfile/temp_control.dbf";
   switch clone tempfile all;
   catalog clone datafilecopy  "+OHIDATA/ohiprd01/datafile/sysaux.274.826592537",
 "+OHIDATA/ohiprd01/datafile/undotbs1.259.826592535",
 "+OHIDATA/ohiprd01/datafile/users.286.826592905",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_fin_ind.283.826592905",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_fin_tab.284.826593409",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_rel_ind.280.826593925",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_rel_tab.264.826592905",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_sys_ind.268.826593409",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_sys_tab.271.826593925",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_ind.278.826593411",
 "+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_tab.272.826593925",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_fin_ind.262.826592539",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_fin_tab.285.826592539",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_rel_ind.267.826593409",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_rel_tab.265.826592905",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_sys_ind.261.826593409",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_sys_tab.279.826592537",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_ind.281.826592905",
 "+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_tab.257.826593925",
 "+OHIDATA/ohiprd01/datafile/ozg_log_ind.275.826593411",
 "+OHIDATA/ohiprd01/datafile/ozg_log_tab.282.826593925",
 "+OHIDATA/ohiprd01/datafile/undotbs2.276.826592539",
 "+OHIDATA/ohiprd01/datafile/tbs_control.dbf",
 "+OHIDATA/ohiprd01/datafile/tbs_control_01.dbf",
 "+OHIDATA/ohiprd01/datafile/tbs_control_02.dbf";
   switch clone datafile all;
}
executing Memory Script

executing command: SET NEWNAME

executing command: SET NEWNAME

renamed tempfile 1 to +ohidata in control file
renamed tempfile 2 to +OHIDATA/ohiprd01/tempfile/temp_control.dbf in control file

cataloged datafile copy
```

datafile copy file name=+OHIDATA/ohiprd01/datafile/sysaux.274.826592537 RECID=1 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/undotbs1.259.826592535 RECID=2 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/users.286.826592905 RECID=3 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_ind.283.826592905 RECID=4 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_tab.284.826593409 RECID=5
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_ind.280.826593925 RECID=6 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_tab.264.826592905 RECID=7
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_ind.268.826593409 RECID=8
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_tab.271.826593925 RECID=9
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_ind.278.826593411 RECID=10
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_tab.272.826593925 RECID=11
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_ind.262.826592539 RECID=12
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_tab.285.826592539 RECID=13
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_ind.267.826593409 RECID=14
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_tab.265.826592905 RECID=15
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_ind.261.826593409 RECID=16
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_tab.279.826592537 RECID=17
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_ind.281.826592905 RECID=18
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_tab.257.826593925 RECID=19
STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_log_ind.275.826593411 RECID=20 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/ozg_log_tab.282.826593925 RECID=21 STAMP=826594490

cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/undotbs2.276.826592539 RECID=22 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/tbs_control.dbf RECID=23 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/tbs_control_01.dbf RECID=24 STAMP=826594490
cataloged datafile copy
datafile copy file name=+OHIDATA/ohiprd01/datafile/tbs_control_02.dbf RECID=25 STAMP=826594490

datafile 2 switched to datafile copy
input datafile copy RECID=1 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/sysaux.274.826592537
datafile 3 switched to datafile copy
input datafile copy RECID=2 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/undotbs1.259.826592535
datafile 4 switched to datafile copy
input datafile copy RECID=3 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/users.286.826592905
datafile 5 switched to datafile copy
input datafile copy RECID=4 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_ind.283.826592905
datafile 6 switched to datafile copy
input datafile copy RECID=5 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_fin_tab.284.826593409
datafile 7 switched to datafile copy
input datafile copy RECID=6 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_ind.280.826593925
datafile 8 switched to datafile copy
input datafile copy RECID=7 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_rel_tab.264.826592905
datafile 9 switched to datafile copy
input datafile copy RECID=8 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_ind.268.826593409
datafile 10 switched to datafile copy
input datafile copy RECID=9 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_sys_tab.271.826593925
datafile 11 switched to datafile copy
input datafile copy RECID=10 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_ind.278.826593411
datafile 12 switched to datafile copy
input datafile copy RECID=11 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_dim_zrg_tab.272.826593925
datafile 13 switched to datafile copy
input datafile copy RECID=12 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_ind.262.826592539
datafile 14 switched to datafile copy
input datafile copy RECID=13 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_fin_tab.285.826592539
datafile 15 switched to datafile copy
input datafile copy RECID=14 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_ind.267.826593409
datafile 16 switched to datafile copy
input datafile copy RECID=15 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_rel_tab.265.826592905
datafile 17 switched to datafile copy
input datafile copy RECID=16 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_ind.261.826593409
datafile 18 switched to datafile copy

```
input datafile copy RECID=17 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_sys_tab.279.826592537
datafile 19 switched to datafile copy
input datafile copy RECID=18 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_ind.281.826592905
datafile 20 switched to datafile copy
input datafile copy RECID=19 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_fact_zrg_tab.257.826593925
datafile 21 switched to datafile copy
input datafile copy RECID=20 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_log_ind.275.826593411
datafile 22 switched to datafile copy
input datafile copy RECID=21 STAMP=826594490 file
name=+OHIDATA/ohiprd01/datafile/ozg_log_tab.282.826593925
datafile 23 switched to datafile copy
input datafile copy RECID=22 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/undotbs2.276.826592539
datafile 24 switched to datafile copy
input datafile copy RECID=23 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/tbs_control.dbf
datafile 25 switched to datafile copy
input datafile copy RECID=24 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/tbs_control_01.dbf
datafile 26 switched to datafile copy
input datafile copy RECID=25 STAMP=826594490 file name=+OHIDATA/ohiprd01/datafile/tbs_control_02.dbf

contents of Memory Script:
{
  Alter clone database open resetlogs;
}
executing Memory Script

database opened
Finished Duplicate Db at 20-SEP-13

RMAN>
```